

Jacob Thomas Sony | Curriculum Vitae

Final-year Mechanical Engineering student at IIT Bombay, India

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RESEARCH INTERESTS

- **Mechanics and Control of Materials, Structures, and Systems** using analytical, computational, and empirical methods
- **Applications:** Environmental and Energy Sustainability, Biomedical Engineering, Healthcare, Robotics, and Automobiles

EDUCATION

Indian Institute of Technology (IIT) Bombay

[2020 - Present]

Dual Degree (B. Tech + M. Tech) in [Mechanical Engineering](#) | CPI: 9.76/10

Minor degrees in [Artificial Intelligence and Data Science](#) & [Systems and Control Engineering](#)

- **Department Rank 2** out of 228 students in the undergraduate Batch of 2020 | Ranked 1st in the integrated program
- Received the **Academic Proficiency (AP)** grade for exceptional performance (top 1%) in 6 key courses, which include -
 - Solid Mechanics
 - Structural Materials
 - Programming for Data Science
 - Biology
 - Introduction to ML

HONORS & ACHIEVEMENTS

- Recipient of the **Institute Academic Award** for outstanding performance | Achieved **10.0** grade in the previous 2 semesters
- Winner of the **Best Paper Award** at the *International Conference on Fluid Flow and Thermal Science 2023* in Portugal
- Among the **top 5** candidates nationwide to be conferred the **2024 RIYA scholarship** by the Ohio State University, USA
- Awarded the prestigious **2023 MITACS Globalink Research Internship** for pursuing fully-funded research in Canada

PUBLICATIONS

- Sakrikar, A., Sony, Jacob. T., Singla, P., and Baranwal, A. "Intelligent Thermal Management of Li-ion Batteries via Temperature Responsive Coolant Flow Control" 4th International Conference on Fluid Flow and Thermal Science (2023)

PROFESSIONAL & RESEARCH EXPERIENCE

Design and Control of Bipedal Walking Robots

[Jul '23 - Present]

Guide(s): [Prof. Vivek Sangwan](#), [IIT Bombay](#) | Master's Thesis

Synopsis: Developing a robotic testbed with a soft foot actuation mechanism for gait initiation and bipedal walking studies

- Designed the **torso-leg system** and **foot actuation mechanism** using DFMA and material/component selection principles
- Fabricated a **customized PCB** in KiCAD to efficiently route signals and streamline the **electro-pneumatic** subsystem
- Developed a virtual **MATLAB environment** that simulates the actuation, swing, and impact mechanics of the biped robot
- Implemented **HZD** and **SLIP control** schemes for accurate torso stabilization and foot placement within **2%** tolerance

Dynamics of QZS Metastructure-based Isolators

[May '24 - Jul '24]

Guide(s): [Prof. Rajendra Singh](#), [Ohio State University](#) & [Dr. Paul Gilmore](#), [Toyota R&D](#)

Synopsis: Examining the efficacy of Quasi-Zero Stiffness (QZS) components for vibration isolation in automotive applications

- Modeled and simulated the dynamics of a **QZS isolator stack-mass** system with **manufacturing-induced** heterogeneity
- Investigated the **excitation response** of the isolator using system linearization, **frequency** and **physical domain** techniques
- Used data-based **parameter tuning** and validated simulation results through vibration experiments, achieving **>85%** match
- Analyzed the effect of **damping parameters** and **operating regime** on transmissibility metrics for improved isolator design

Mechanics of Marine-conditioned FRP-reinforced Concrete Columns

[May '23 - Jul '23]

Guide(s): [Prof. Shamim Sheikh](#), [University of Toronto](#) & [Prof. Nitesh P. Yelve](#), [IIT Bombay](#)

Synopsis: Exploring alternatives to steel for sustainable concrete structures in extreme conditions intensified by climate change

- Involved in the **instrumentation** and **compression** testing of 20+ uniquely conditioned steel & FRP-reinforced columns
- Played a pivotal role in **characterization** and **fabrication** of the rebars, concrete cylinders and other necessary components
- Analyzed the effect of reinforcing **material** and **architecture** on **structural confinement** in the post-peak and failure phases
- Examined cross-sections of failed specimens using **microscopy** and **imaging** methods to study microstructural features

Remote Monitoring and Estimation of Upper Limb Kinematics

[Jan '24 - May '24]

Guide(s): [Prof. Darshan Shah, IIT Bombay](#) | Research & Development Project

Synopsis: Investigating strategies to aid in the cost-effective rehabilitation of people with motor disabilities in the Indian setting

- Laid the foundation for a **data acquisition** and analysis pipeline to monitor the **rehabilitation** process of stroke patients
- Performed **multi-DoF experiments** on fabricated hinges and human upper limbs for **state estimation** using **IMU** data
- Deployed algorithms that utilize the **estimated IMU quaternion** data to extract **kinematic parameters** like joint angles

Optimal Control of Battery Thermal Management System

[Jan '23 - Dec '23]

Synopsis: Devising novel and efficient thermal management schemes for batteries in electric vehicles (EVs), ensuring longevity

- Proposed alternate models based on **equilibrium temperature** to quantify the **thermal degradation** of Li-ion batteries
- Formulated a **nonlinear discrete optimization** problem to optimize the coolant flow rate that reduces the operating cost
- Obtained results which indicated up to **5% cost reduction**, exploiting the trade-off between degradation and pumping power

TECHNICAL EXPERIENCE

SLAM Engineer | IIT Bombay Racing

[2020 - 2022]

A cross-functional team of 70+ students working on designing and manufacturing race cars for 'Formula Student' competitions
'IIT B Racing Driverless' is the first Indian team to participate in the *Formula Student AI (UK)* competition at Silverstone

- Involved in the development of IIT Bombay Racing's driverless race car, incorporating end-to-end **simulation** and **testing**
- Implemented some of the earliest in-house simulator-based testing platforms, like **track generators** and **state visualizers**
- Created and tested initial versions of the **SLAM module** for accurate localization of the vehicle and mapping of the track
- Designed a pipeline for **sensor fusion** and **state estimation** utilizing **odometry** and **LiDAR** along with vehicle dynamics
- Secured **3rd** place in the 'FSEV 2021 Software and Intelligence Integration' competition, sponsored by **Ather Energy**

KEY PROJECTS

Fracture Mechanics of 'Single Edge Notch Tension' Specimens

[Jan '22 - May '22]

Guide(s): [Prof. Krishna N. Jonnalagadda](#) | Experimental Solid Mechanics

- Fabricated aluminium **SENT** specimens of different **crack lengths** in the lab and conducted **tensile tests** in the UTM
- Applied **regression** methods on the **load-deflection** data to obtain the relation between **fracture load** and **crack length**

Analysis of Tsai-Wu Failure Criterion for Composites

[Jan '22 - May '22]

Guide(s): [Prof. Nitesh P. Yelve](#) | Design of Fiber-Reinforced Composites

- Implemented a **MATLAB solver** to obtain the failure envelope of a **Graphite/Epoxy** laminate given the design parameters
- Created an **interactive display** with the ability to vary parameters to **design for failure** under biaxial loading conditions

Deformation Mechanisms of Crystals

[Jan '23 - May '23]

Guide(s): [Prof. Alankar Alankar](#) | Mechanisms of Crystal Plasticity

- Studied the theoretical, experimental, and computational research work done on mechanisms like **plastic slip** and **twinning**
- Analyzed dislocation paradigms such as the **P-N model** used for strain-hardening in **TWIP steel** and **high-entropy alloys**
- Compiled the various results in the form of a comprehensive **review paper** based on the extensive literature study of papers

Study of Short-pulse Laser Heating of Metals

[Jan '22 - May '22]

Guide(s): [Prof. Ramesh Singh](#) | Laser Manufacturing

- Performed **transient thermal FEA** simulations of **laser heating** of an aluminium block to study features like melt depth
- Modeled the laser engraving process using **analytical** principles of **heat transfer**, and compared the results with simulation
- Carried out **experiments** to study the effect of parameters like **power** on performance aspects such as penetration depth

Kinematic and Dynamic Analysis of Ornithopter Wing

[Jan '24 - May '24]

Guide(s): [Prof. Anirban Guha](#) | Computer Aided Simulation of Machines

- Created the **CAD** model of a 6-link **flapping-cum-flexing** wing mechanism in **Fusion 360** with optimal flight attributes
- Conducted **multibody simulations** in **ADAMS** to study the mechanics under different **loading** and **actuation** conditions
- Performed **design optimization** of the mechanism and **sensitivity analysis** on geometric and joint clearance parameters

Pose Estimation and Feedback Control of Differential Drive Robots

[Jul '23 - Dec '23]

Guide(s): [Prof. Leena Vacchani](#) | Embedded Control and Robotics

- Employed the **Extended Kalman Filter** algorithm for **localization** and pose estimation using **motion** and **sensor** models
- Utilized a smooth, **Lyapunov**-based non-linear feedback control for **full-state stabilization** and **steering** of the vehicle
- Incorporated **adaptive gain tuning** into the steering control, enabling successful **path navigation** with **< 2% deviation**

Reinforcement Learning for System Swing-up and Stabilization

[Jan '24 - May '24]

Guide(s): [Prof. Leena Vacchani](#) | *Intelligent Feedback and Control*

- Created a **training environment** in MATLAB with a suitable **reward function**, as well as **action** and **observation** spaces
- Designed a continuous **Deep Deterministic Policy Gradient Agent** with proper **Actor** and **Critic** network architectures
- Trained the **model-free agent** to swing up and stabilize an **inverted pendulum** subject to torque and other constraints
- Accomplished desirable results from different initial conditions with $< 5\%$ **overshoot** while being **robust** to **sensor noise**

Deep Learning based Trajectory tracking for Robotic Manipulators

[Jan '24 - May '24]

Guide(s): [Prof. Amit Sethi](#) | *Neural Networks and Deep Learning*

- Generated **training data** consisting of **end-effector positions** and **joint orientations** in Python using forward kinematics
- Implemented different models such as **Recurrent Neural Networks (RNNs)** and compared their **tracking performance**
- Attained an **average tracking coefficient** of **0.986** across various standard trajectories with **smooth joint angle variation**

POSITIONS OF RESPONSIBILITY

Core Team Member | Electronics and Robotics Club, IIT Bombay

[2021 - 2022]

A 10-member team responsible for catering to the nation's electronics and robotics enthusiasts via events & competitions

- Participated in the **e-Yantra Robotics** Competition, involving the development of an **autonomous milk delivery** robot
- Collaborated with the educational robotics firm **RoboBox** to organize a **line-follower** workshop for **500+** school students
- Worked with the tech clubs of top IITs under '**All IIT Robotics Association**' to conduct a **pan-India** Robotics Challenge
- Moderated the **Control Theory Bootcamp** which focused on **PID** and **LQR** control for different systems in **MATLAB**
- Organized sessions on simulating **mobile robots** in **ROS** and **Gazebo** for implementing **obstacle avoidance** algorithms
- Mentored students regarding **mechanical & electrical** design aspects of a **multi-DOF robotic arm** in the flagship event

Teaching Assistant | Department of Mechanical Engineering, IIT Bombay

[Jul '24 - Present]

Instructor: [Prof. Vivek Sangwan](#) | Course: *Microprocessors and Automatic Control*

- Responsible for guiding **100+** third-year students as an undergraduate TA through doubt resolution and tutorial sessions
- Played an instrumental role in managing student **attendance** in lectures as well as **evaluating** answer scripts of exams

Teaching Assistant | Department of Mathematics, IIT Bombay

[Jan '24 - May '24]

Instructor: [Prof. Sudarshan R. Gurjar](#) | Course: *Introduction to Numerical Analysis*

- Served as a teaching assistant for **50+** undergraduates, conducting weekly tutorial sessions and grading answer sheets
- Assisted the instructor regularly in **formulating** problems and **proofreading** solutions for tutorials throughout the semester

Mentor | Department Academic and Career Mentorship Programme

[2022 - Present]

A department initiative aimed at ensuring effective academic and career guidance & mentorship for sophomores and juniors

- Responsible for guiding students in academics, internships, and higher education through **live help sessions** and **seminars**
- Provided **personalized assistance** to students in preparing **CVs** and **Statements of Purpose** for internship applications
- Contributed to the department blog through carefully outlined **reviews** on different **department** and **institute electives**

Student Volunteer | National Service Scheme, IIT Bombay

[2020 - 2021]

The largest on-campus student volunteer organization, dedicated to social service through educational & technical initiatives

- Involved in **80+** hours of volunteering to promote awareness about **sustainable social development** through multimedia
- Created **educational videos** on different **social, economic** and **environmental** concerns for the official Youtube channel

MISCELLANEOUS

COURSEWORK	• Mechanical - Solid Mechanics, Structural Materials, Thermodynamics, Machine Design, Fluid Mechanics, Operations Research, Manufacturing Processes, Engineering Dynamics • Others - Calculus, Linear Algebra, Differential Equations, Linear & Non-Linear Systems, Intelligent Feedback and Control, Embedded Control and Robotics, Numerical Analysis
TECHNICAL SKILLS	• Languages & Programming - C++, Python, MATLAB, Octave, SQL, G-Code, $\LaTeX$ • Software Packages - Fusion 360, SimScale, ADAMS, ArduinoIDE, KiCAD, ROS, Gazebo
EXTRACURRICULARS	• Secured 4th position nationally in RaScionix 2020 organized by the Chemistry Club • Appointed as the Vice Head Boy for the academic year 2018-19 based on all-round profile • Enthusiastic fan of motorsports like F1 & MotoGP Hobbyist Animator & 3D Modeler